UE23CS320B – Capstone Project Phase – 2 ESA

Project Progress Review #3

Project Title : ***DreamDrive***: Proactive Multi-Future Prediction for Safe Autonomous Navigation in Chaotic Mixed Traffic

Project ID : PW26_CGM_04

Project Guide : Prof. Chitra G M

Project Team : Idhant Singh, Rishav Ghosh, Shagal Verma, Shourya Rai

PES UNIVERSITY

Outline

- Abstract and scope.
- Suggestions from phase-1
- Design Constraints, Assumptions, Dependencies
- Data Exploration
- Architecture
- Use Case Diagram
- Benchmarks-1
- Conclusion and Suggestions from Phase 2 Review 1
- Benchmarks-2
- Observations and Conclusion from Phase 2 Review 2
- Additional Info and References

Abstract and Scope

- Current autonomous driving systems rely primarily on ***reactive decision-making***, predicting a single, most-likely future path for surrounding agents.
- ***This approach fails in chaotic, unstructured traffic environments*** such as those found in India, where pedestrians, rickshaws, and two-wheelers behave unpredictably.
- There is therefore a pressing need for a ***proactive prediction model*** that does not depend on a single deterministic forecast, but instead *simulates multiple possible futures*.
- ***DreamDrive*** addresses this challenge by enabling autonomous vehicles to anticipate risks before they occur, ensuring safer navigation in mixed-traffic environments.

Suggestions from Phase-1

- **Suggestion:**

  Narrow the project scope by explicitly defining and prioritizing a limited set of traffic participants and object categories that have the highest impact on safety in chaotic mixed-traffic environments.

- **Feasibility & Current Progress:**

  Scoped Focus:
  The current phase is limited to vehicle-to-vehicle interactions to ensure reliable model development.
  Targeted Dataset Selection:
  Only vehicle-dominant scenes from the distributed dataset are used for training and evaluation.
  Phased Extension:
  The pipeline is designed to later incorporate pedestrians and other agents without redesign.

Design Constraints

- **Dataset Constraints:**
  Limited availability of large-scale, well-annotated Indian chaotic traffic datasets; significant behavioral and structural domain gap between Indian and Western datasets.

- **Computational Constraints:**
  1. Multi-future prediction (K trajectories per agent) increases inference time and memory requirements, limiting real-time scalability.
  2. Best-of-K training further increases training cost due to **oracle** selection.

- **Simulation Constraints:**
  1. CARLA does not natively model all Indian traffic behaviors (lane-less driving, informal overtaking, rickshaw dynamics), requiring approximations.
  2. Traffic density and heterogeneity must be artificially scripted in CARLA to approximate Indian scenarios.

Assumptions

- Traffic agents exhibit locally consistent motion patterns over short prediction horizons.
- Past trajectory history and local spatial context are sufficient to infer near-future behavior.
- Agent detection and trajectory annotations are assumed to be reasonably accurate.
- Road geometry and environmental context remain static during the prediction window.

Dependencies

- **Datasets: IDD-3D** for chaotic traffic; **Argoverse / Waymo / nuScenes** for structured traffic comparison.
- **Models:** Availability of baseline models such as **CV, ST-GAT, MFP and Trajectron++**.
- **Simulation Platform: CARLA** simulator for scenario-based evaluation.
- **Frameworks: PyTorch** and associated deep learning libraries like TorchVision for model development. CUDA-enabled GPU for multi-agent and multi-future training.

Data Exploration

Dataset details:

Source:

- The project utilizes the **IDD-3D (India Driving Dataset - 3D)**.

Data Characteristics:

- It consists of raw 3D annotations (typically derived from LiDAR or stereo-camera setups) focused on unstructured Indian driving scenarios.

Link: India Driving Dataset

Data Exploration

Size and Type of Data

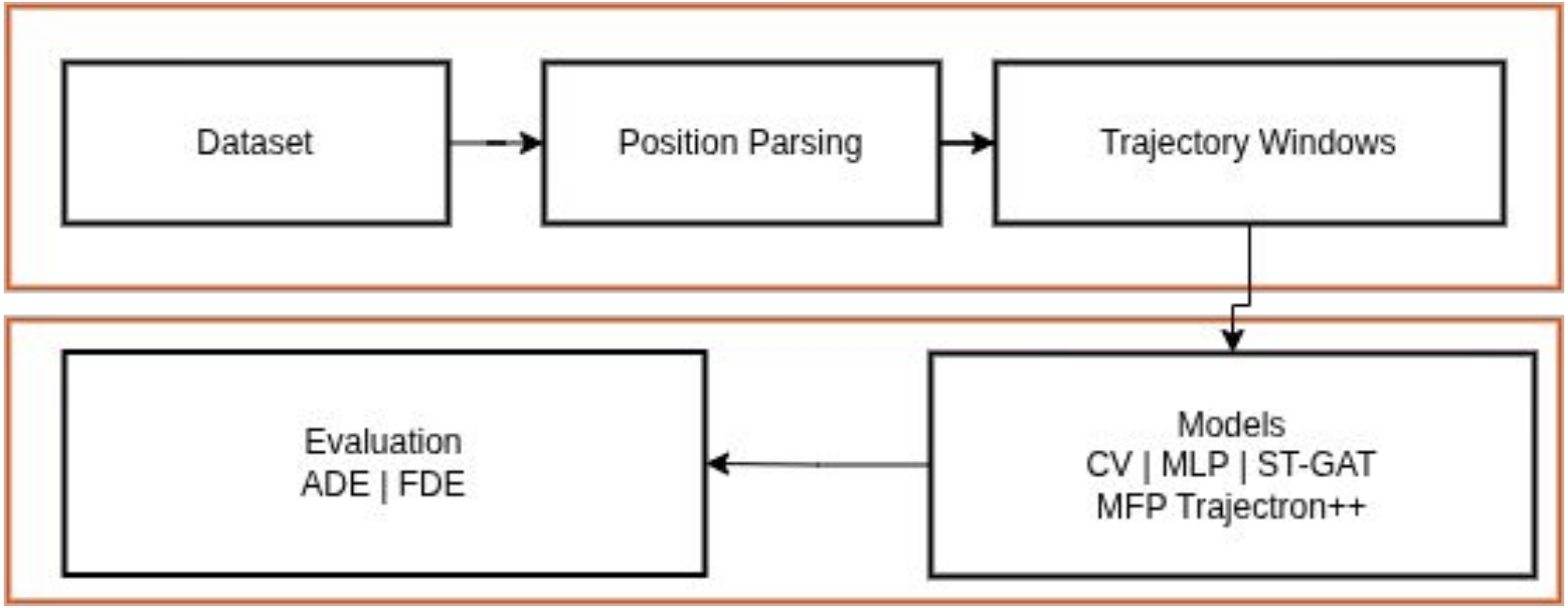
- **Data Type:** 3D Object Detection and Trajectory data. The raw input consists of **JSON label files** per frame, containing oriented bounding box (OBB) information ($psr.position.x, y, z$).
- **Scale (Processed):**
  - **Single-Agent Windows:** N = 4,821 trajectory windows.
  - **Multi-Agent Scenes:** Exported as 138 validation scenes (based on the 80/20 split mentioned in the Trajectron++ section).
- **Format:** The pipeline converts raw JSONs into structured **CSV** (positions), **NPZ** (numpy binaries for training), and **Pickle** (for Trajectron++) formats.

Data Exploration

Time Period / Temporal Resolution

- **Sampling Rate:** The data is processed at **10 FPS** (dt = 0.1s).
  - *Note:* A critical correction was identified during the Trajectron++ evaluation where the default dt = 0.4s (2.5 FPS) was corrected to 0.1s to match the actual IDD-3D capture rate.
- **Window Configuration:**
  - **Observation Length:** 20 frames (2.0 seconds).
  - **Prediction Horizon:** 60 frames (6.0 seconds).
  - **Stride:** 3 frames.

Architecture



Architecture

Data Preparation and Trajectory Construction:

- Raw traffic datasets are parsed to extract precise agent position information, which is then segmented into fixed-length trajectory windows suitable for temporal modeling.

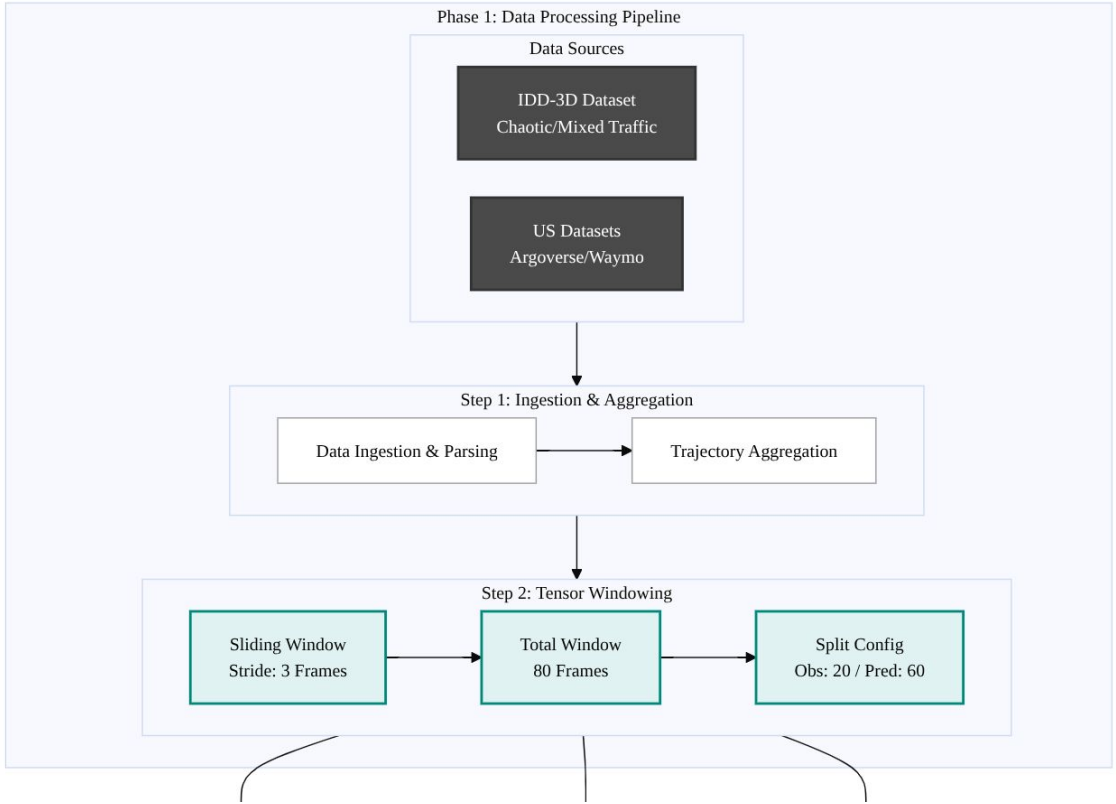
Model Training and Prediction:

- Multiple trajectory prediction models, ranging from physics-based to interaction-aware learning architectures, are trained on the constructed trajectory windows to generate future motion predictions.

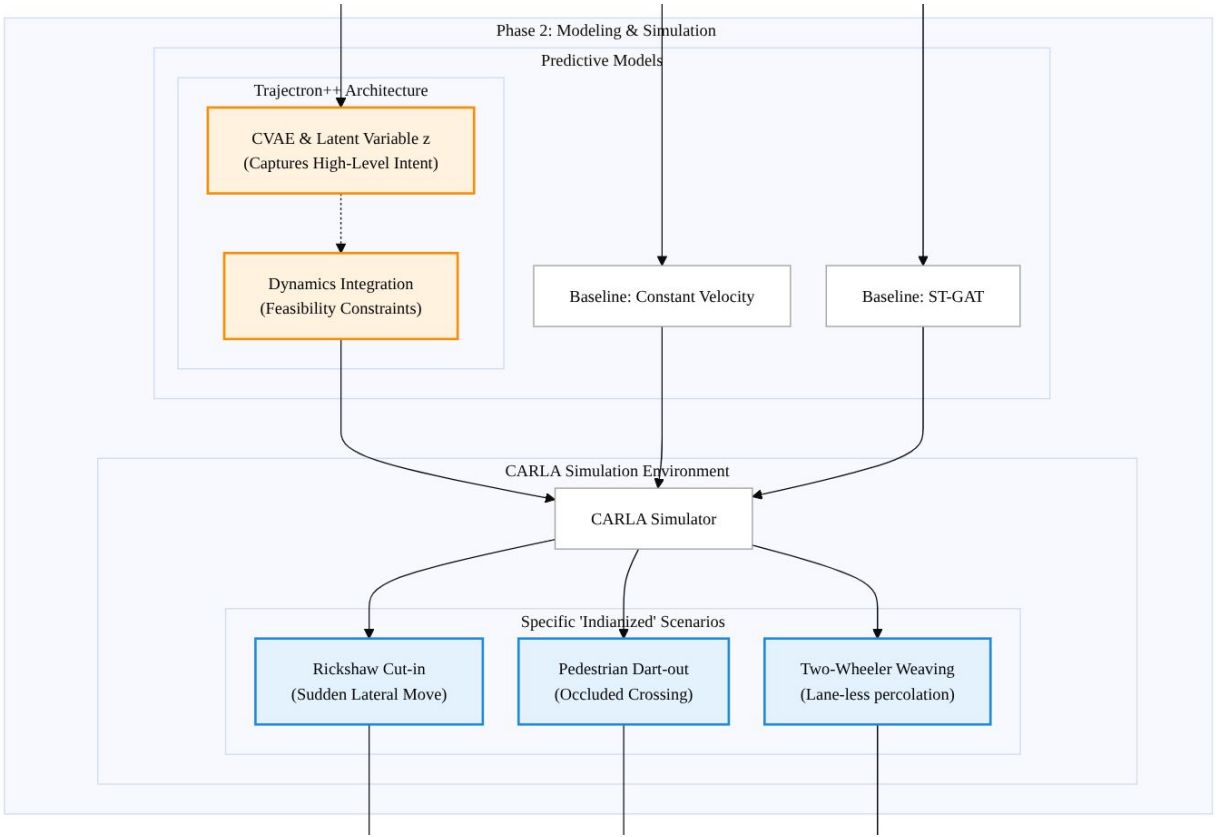
Quantitative Evaluation:

- Model performance is systematically evaluated using standard prediction accuracy metrics, namely Average Displacement Error (ADE) and Final Displacement Error (FDE), enabling objective comparison across approaches.

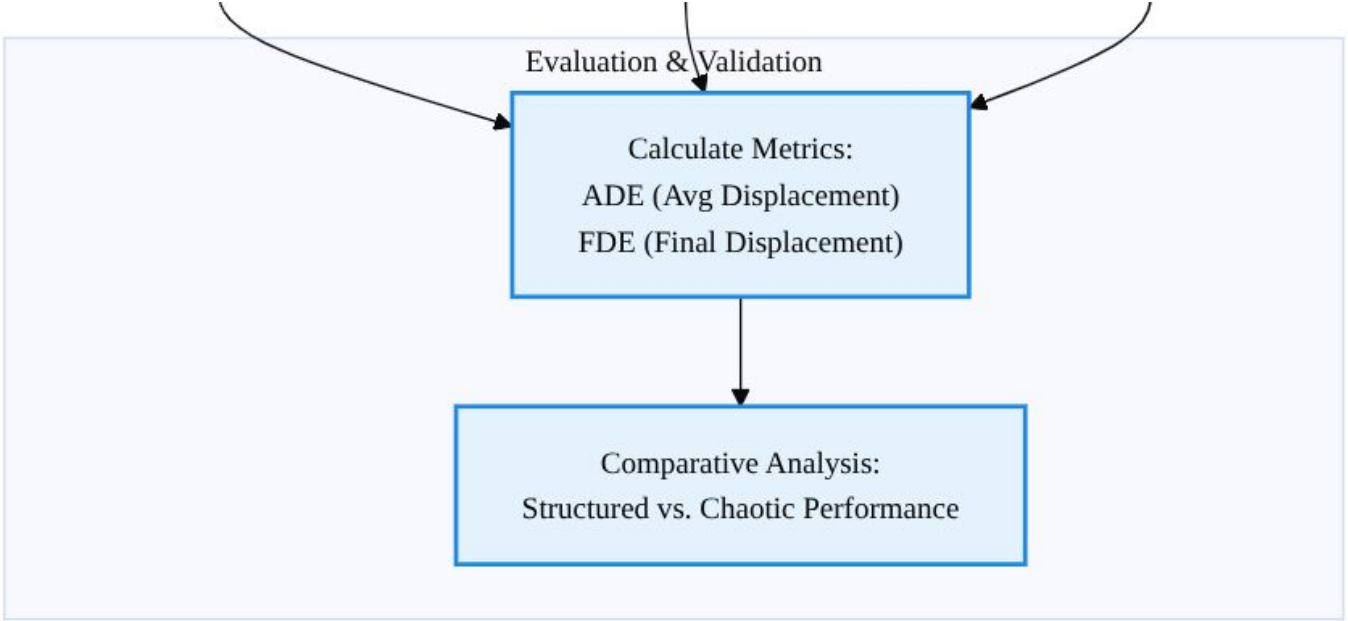
Architecture



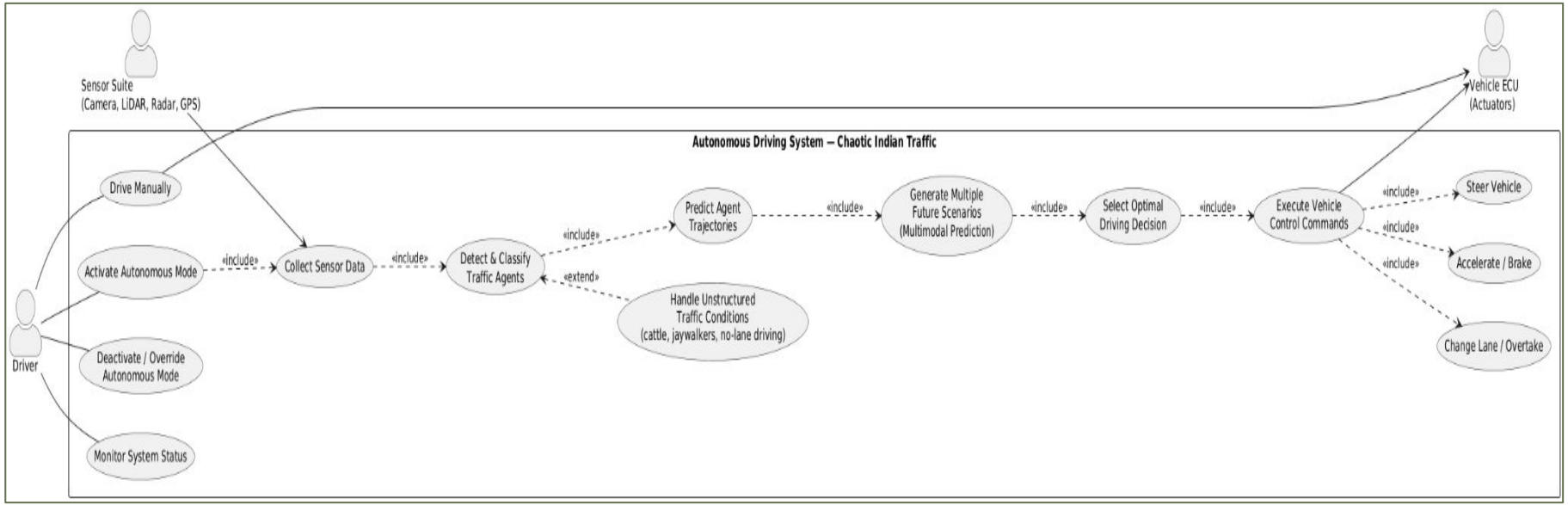
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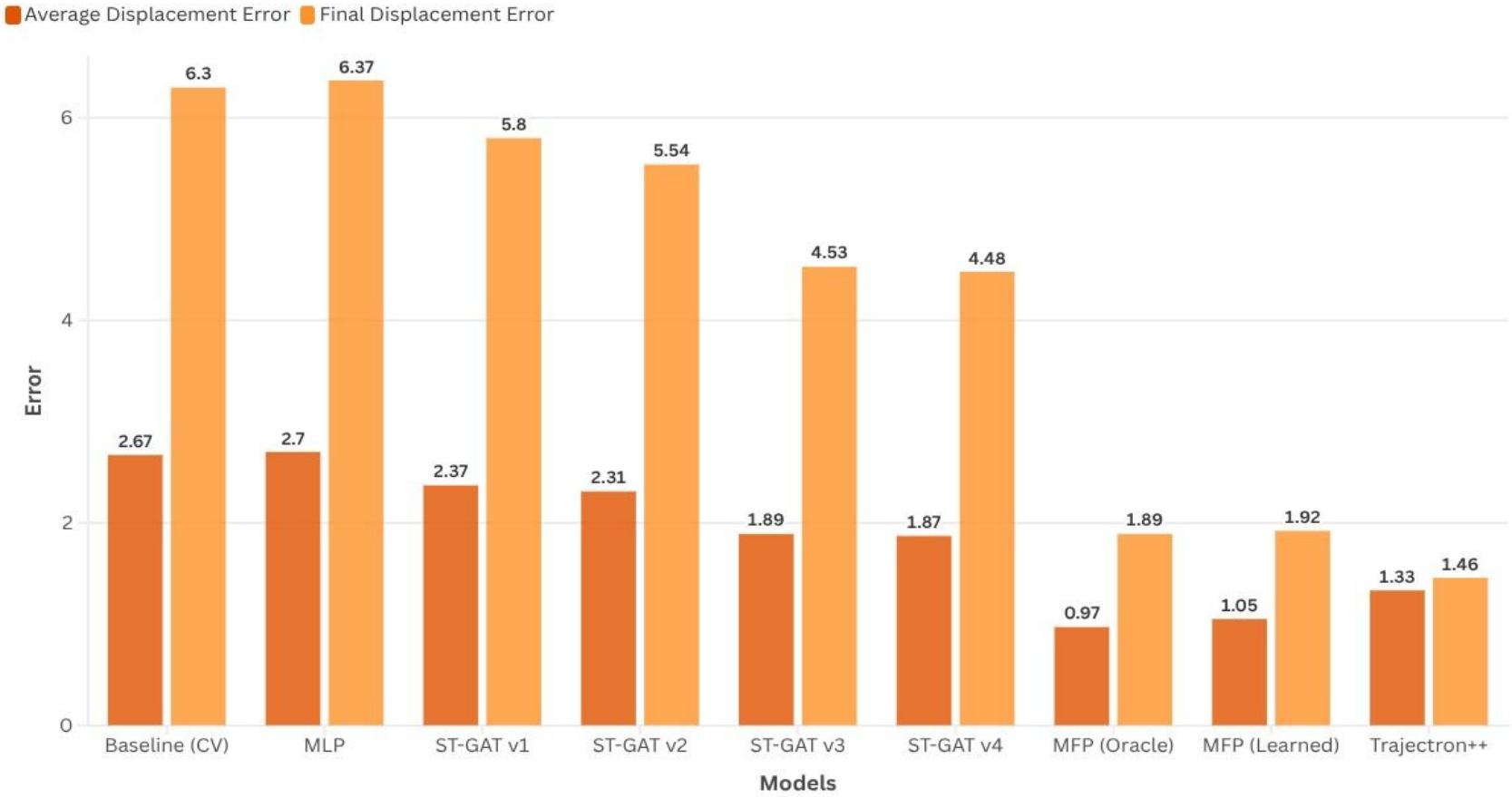
Architecture



Use Case Diagram



Comparative Analysis of ADE/FDE Across Prediction Architectures



Conclusion from Phase 2 Review 1

- **Worst Performance:** MLP
- **Best ADE:** MFP (Oracle) {Best at trajectory shape & uncertainty coverage (**low ADE**)}
- **Best FDE:** Trajectron++* {Best at final goal selection (**low FDE**)}
- **Best Balanced Realistic Model:** MFP(Learned)

* : The ADE and FDE error values obtained for Trajectron++ could potentially be further optimized. However, due to time constraints, exhaustive hyperparameter tuning and extended training were not feasible, and therefore the reported results do not represent the model's fully optimized performance.

Suggestions from Phase 2 Review 1

During Phase 2, we implemented the **Trajectron++** model; however, due to **time constraints**, we were unable to fully optimize its performance.
Limitations included:

- Incomplete hyperparameter tuning
- Limited training time and experimentation
- Suboptimal handling of Indian traffic-specific behaviors

Based on the panel's feedback, we planned to:

- Perform **extensive hyperparameter tuning** and longer training cycles
- Better adapt the model to **heterogeneous and dense Indian traffic scenarios**

Suggestions from Phase 2 Review 1

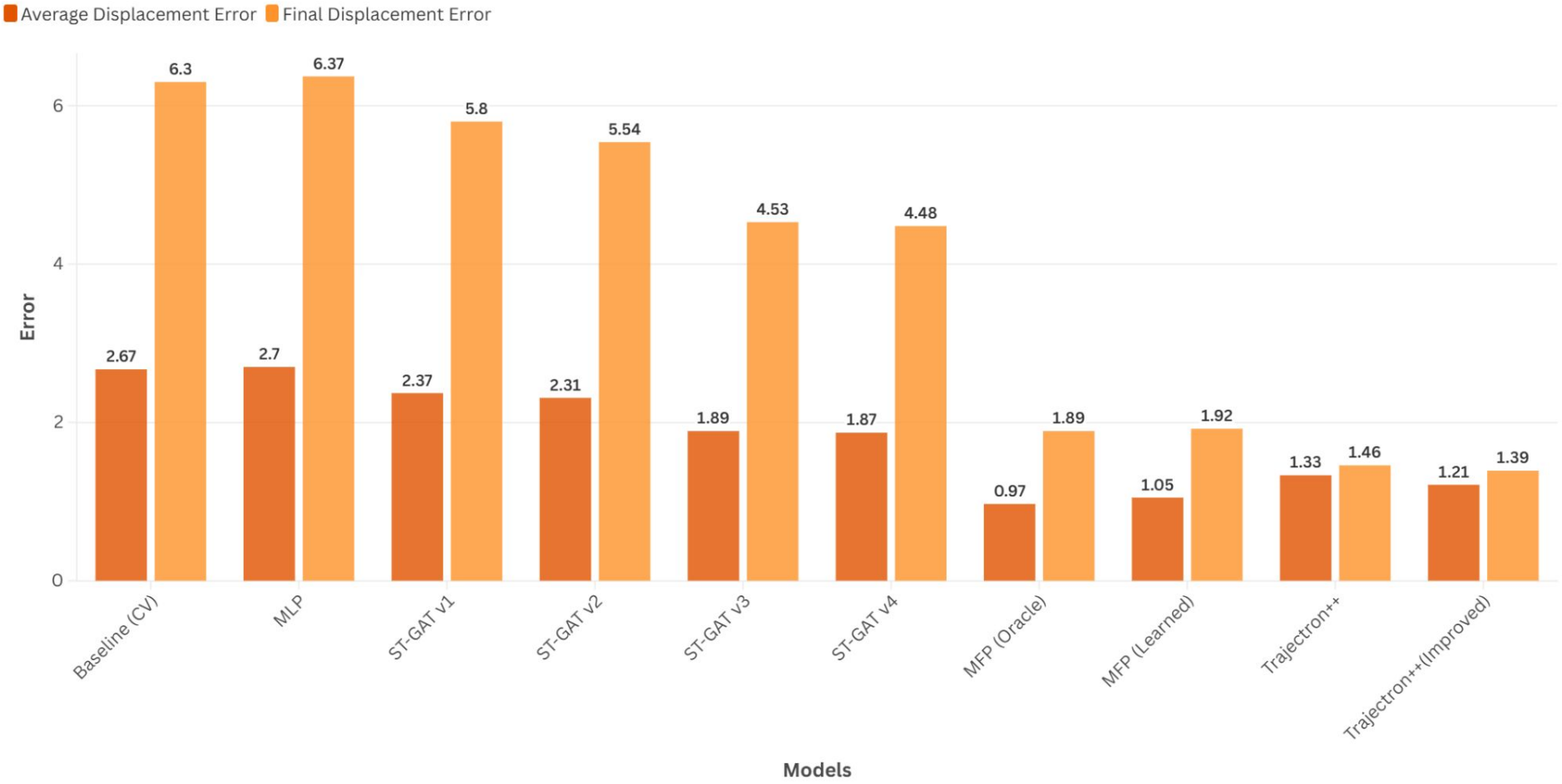
The panel recommended improving model performance on **rare but critical collision scenarios**, which are underrepresented in real-world datasets.

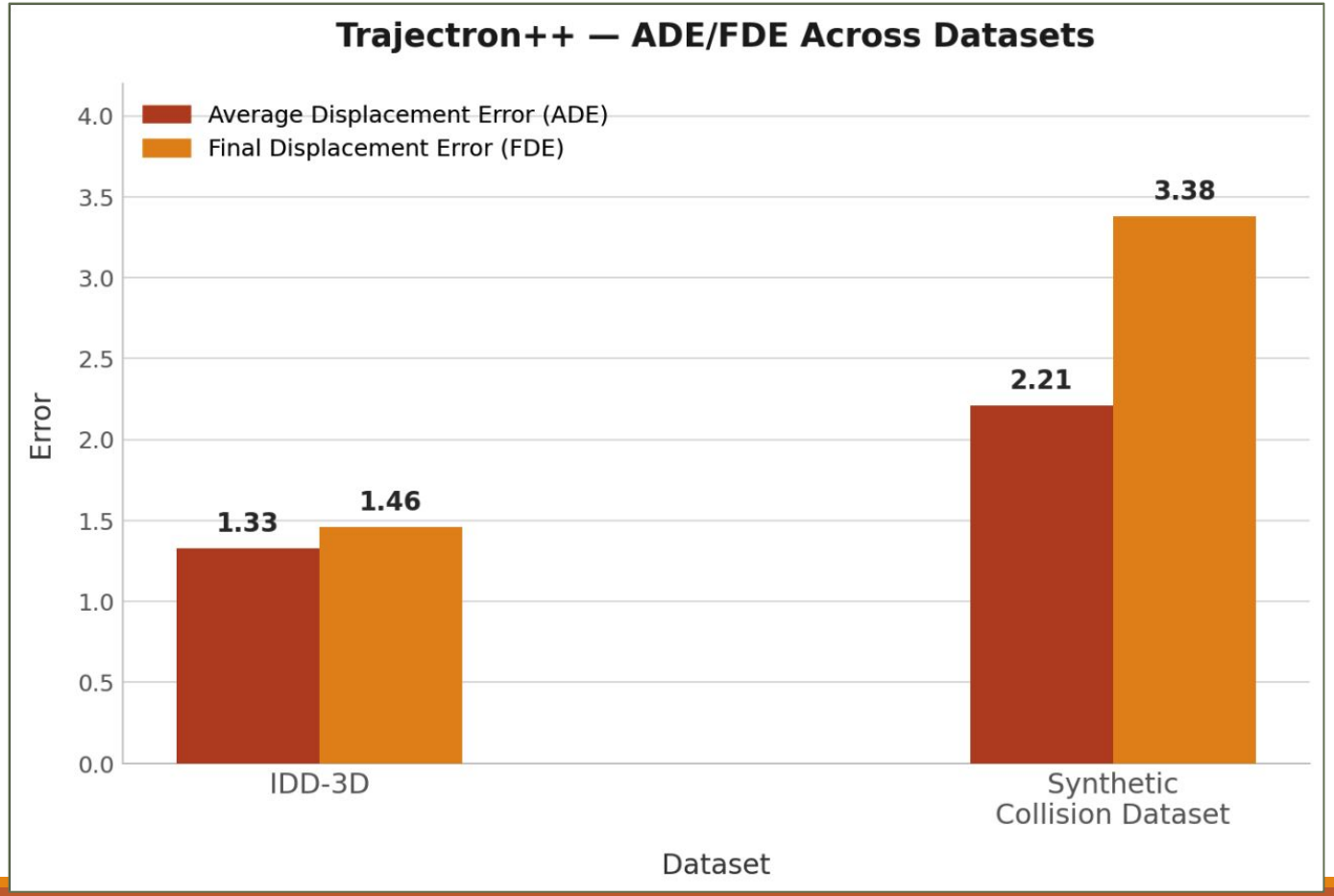
To address this, we are exploring **synthetic collision datasets** to augment training data
Planned approach:

- Use synthetic data to **augment collision and near-collision cases**
- Improve model's ability to **predict risky trajectories and edge-case behaviors**

Goal:
Increase robustness and safety of trajectory prediction by **training on collision-rich synthetic scenarios** that are difficult to capture in real-world data.

Comparative Analysis of ADE/FDE Across Prediction Architectures




Trajectron++ — ADE/FDE Across Datasets
Average Displacement Error (ADE)
Final Displacement Error (FDE)
Error
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
1.33
1.46
2.21
3.38
IDD-3D
Synthetic Collision Dataset
Dataset

Conclusion from Phase 2 Review 2

Key Finding: Trajectron++ shows **66% higher ADE** and **131% higher FDE** on collision data. A planner built on these predictions will inherit these failures.

- Any planner relying on a prediction model is only as safe as the predictions it receives; poor collision-scenario forecasting directly limits collision avoidance capability.
- In the next phase, we aim to fine-tune Trajectron++ on synthetic collision data to close this performance gap.
- We will evaluate whether improved predictions translate to measurable safety gains when integrated into a framework.
- Once prediction quality is validated, we will identify an architecture best suited to unstructured, collision-prone traffic conditions, this will serve as the foundation on which the remainder of the project is built.

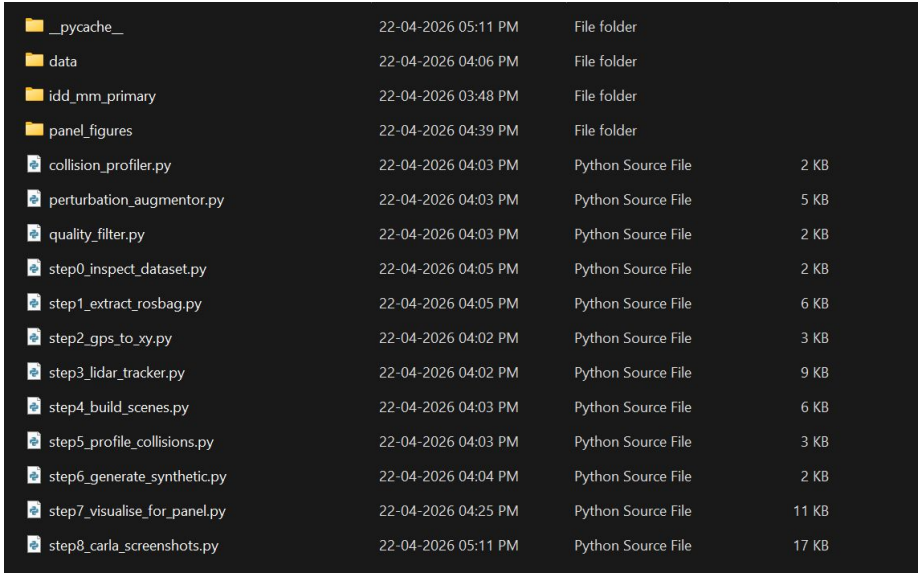
Observations

While training we noticed that our data does not have enough chaotic scenes, so we plan to train on LLM generated synthetic chaotic data as well.

[Enhancing Autonomous Driving Safety with Collision Scenario Integration]

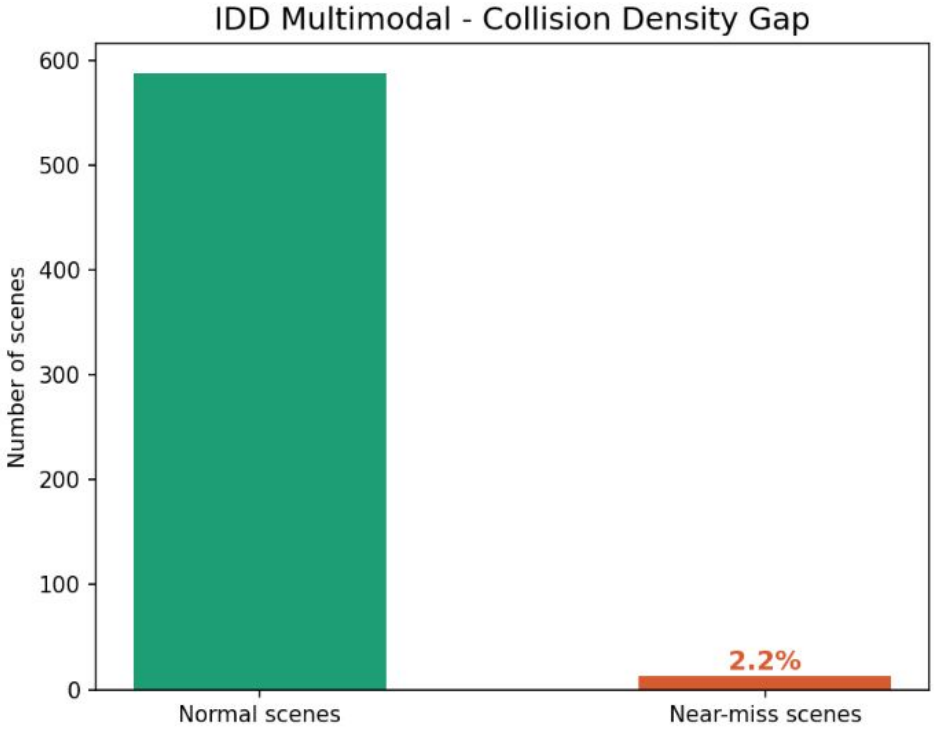
Code Part

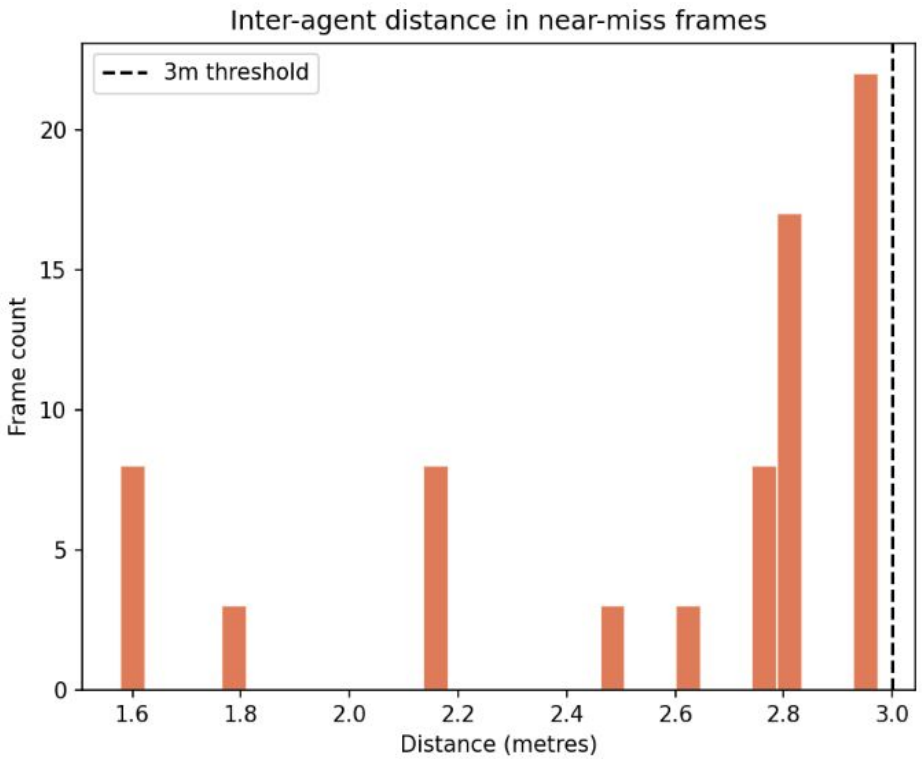
Github:-
https://github.com/im-mrnull/capstone_dreamDrive



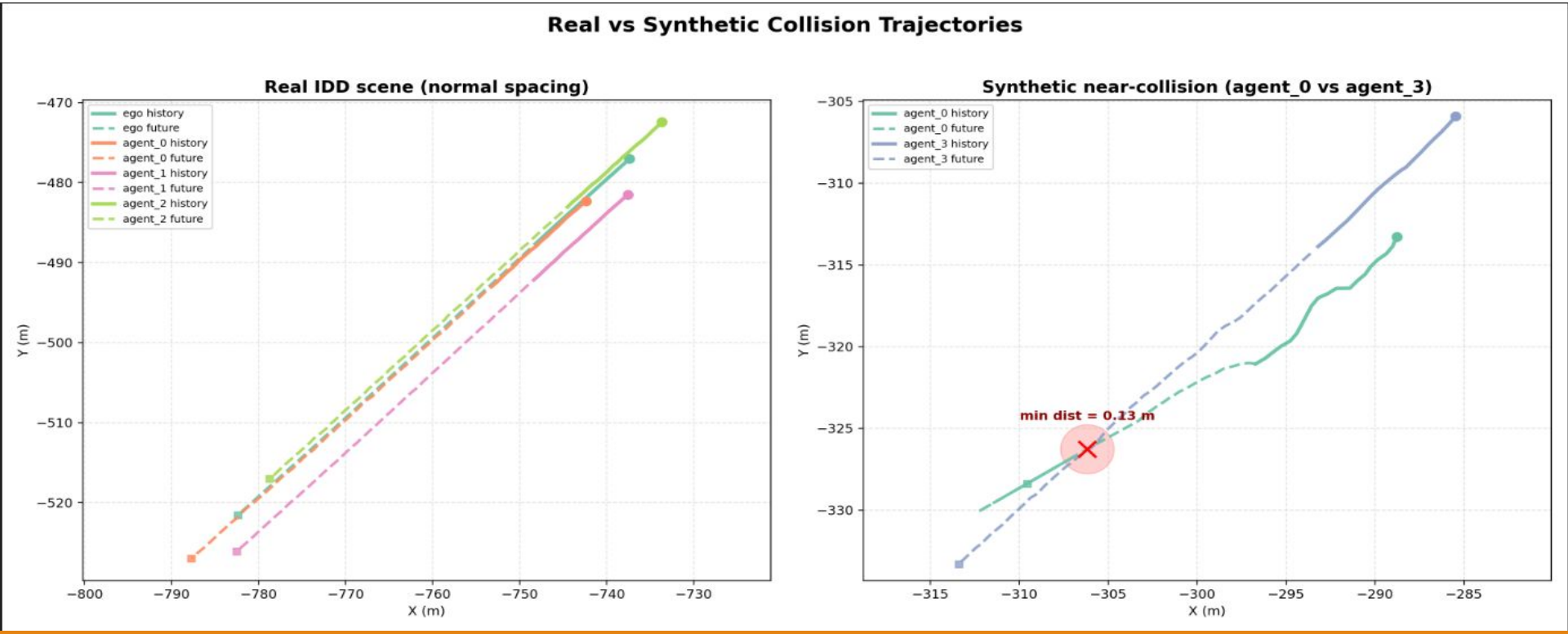
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step0_inspect_dataset.py	22-04-2026 04:05 PM	Python Source File	2 KB
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step6_generate_synthetic.py	22-04-2026 04:04 PM	Python Source File	2 KB
step7_visualise_for_panel.py	22-04-2026 04:25 PM	Python Source File	11 KB
step8_carla_screenshots.py	22-04-2026 05:11 PM	Python Source File	17 KB

Collision Density

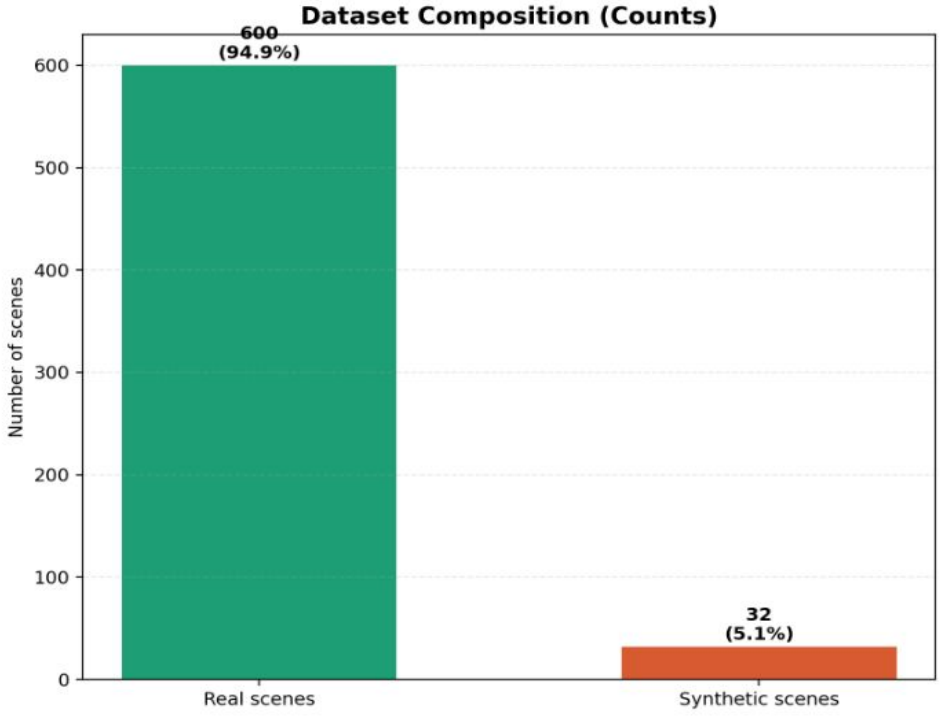


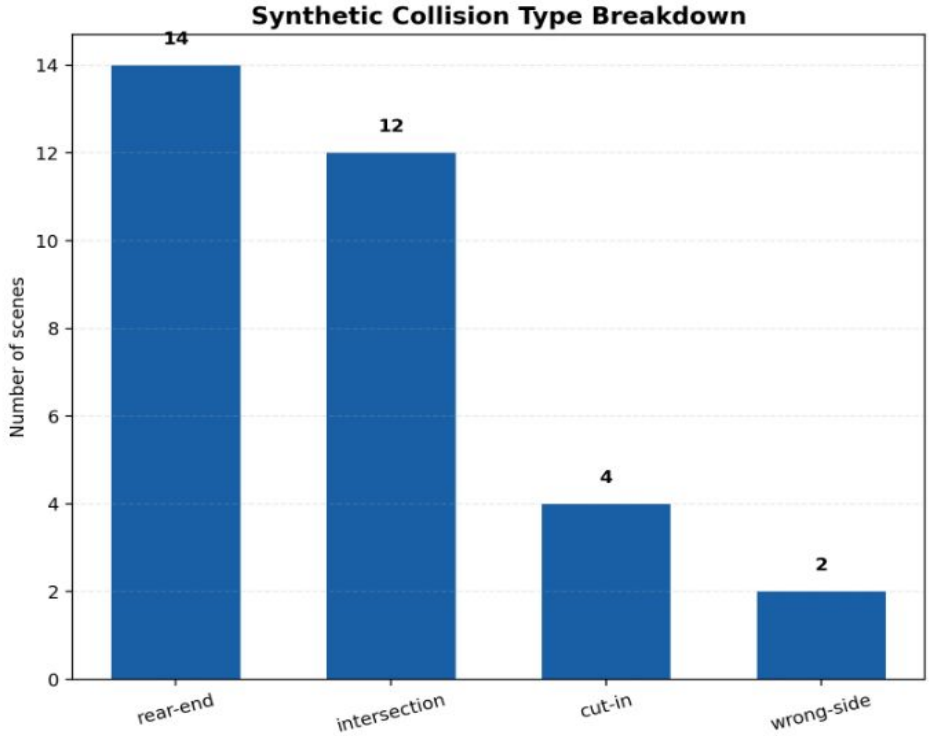


Trajectory Comaparison

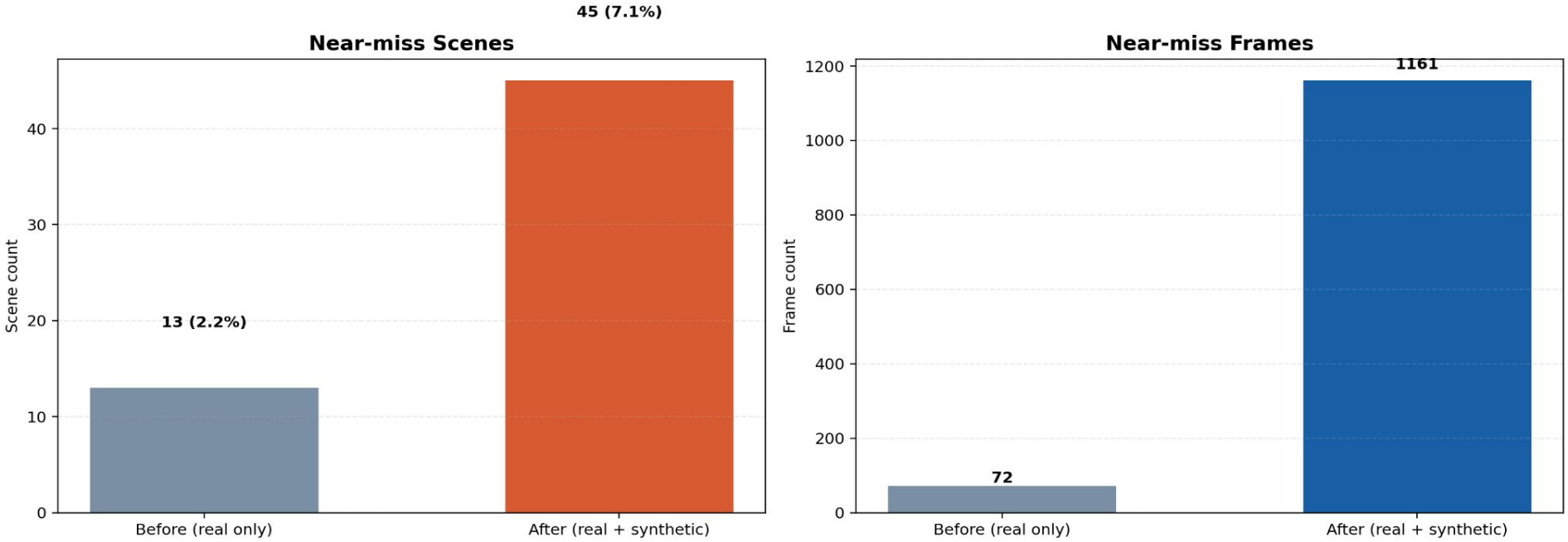


Dataset Composition





Change in Near Miss Augumentation



Carla Simulation Image



Carla Simulation Image



Conclusion

- Single-path prediction is insufficient for chaotic traffic.
- Collision scenarios remain underrepresented in real datasets.
- Model performance drops significantly in safety-critical cases.
- Synthetic data + multi-future prediction improves robustness.
- Enables safer autonomous driving in real-world conditions.

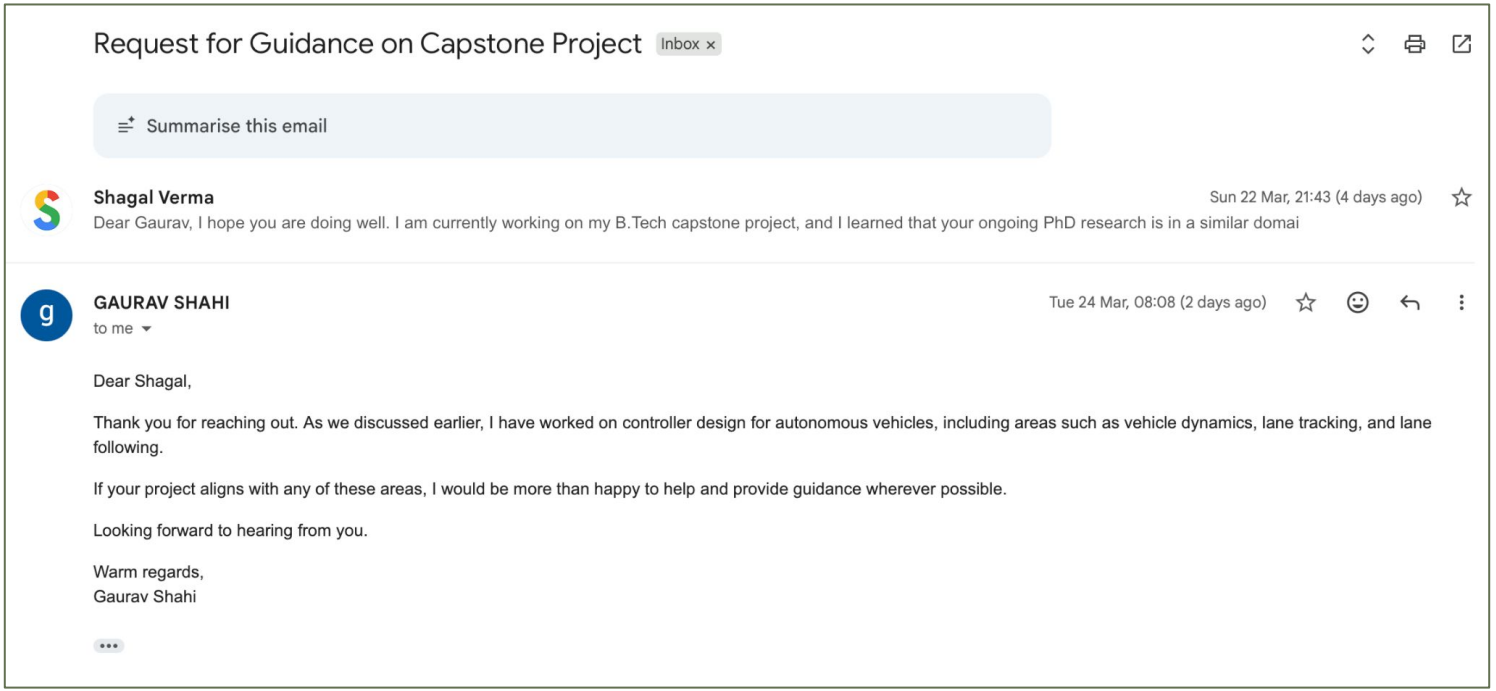
Future Work

- Fine-tune Trajectron++ on collision-rich synthetic datasets.
- Improve modeling of heterogeneous agents (pedestrians, two-wheelers, rickshaws).
- Enhance realism of simulations to better reflect Indian traffic behavior.
- Integrate prediction with planning to evaluate real-world safety impact.
- Scale dataset generation with diverse scenarios and edge cases.

Additional Information

- Contacted an experienced engineer and researcher currently pursuing PhD research in a domain closely aligned with our project.
- His expertise includes areas such as autonomous vehicle control systems, vehicle dynamics, and lane tracking/trajectory-related problems.
- We plan to actively seek his guidance throughout the project lifecycle for:
  - Design validation
  - Technical direction
  - Problem-solving and optimization strategies
- This mentorship is expected to enhance the robustness, accuracy, and real-world applicability of our solution.

Additional Information



Request for Guidance on Capstone Project Inbox ×

Summarise this email

Shagal Verma Sun 22 Mar, 21:43 (4 days ago)

Dear Gaurav, I hope you are doing well. I am currently working on my B.Tech capstone project, and I learned that your ongoing PhD research is in a similar domai

GAURAV SHAHI Tue 24 Mar, 08:08 (2 days ago)

to me

Dear Shagal,

Thank you for reaching out. As we discussed earlier, I have worked on controller design for autonomous vehicles, including areas such as vehicle dynamics, lane tracking, and lane following.

If your project aligns with any of these areas, I would be more than happy to help and provide guidance wherever possible.

Looking forward to hearing from you.

Warm regards,
Gaurav Shahi

References

Provide references pertaining to your research according to IEEE format.

Example:

[1] G. Eason, B. Noble, and I. N. Sneddon, “On certain integrals of Lipschitz-Hankel type involving products of Bessel functions,” Phil. Trans. Roy. Soc. London, vol. AK47, pp. 529–551, April 1955. *(references)*

Thank You!